

Yuang Zhang

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Research Interests

Algorithmic game theory, mechanism design, information elicitation, strategic auditing, peer prediction, and data markets.

Education

George Mason University, Fairfax, VA

2023–present

Ph.D. Student in Computer Science; GPA: 4.0/4.0.

Advisor: Fang-Yi Yu.

The George Washington University, Washington, DC

Sep. 2021–May 2023

M.S. in Computer Science; GPA: 4.0/4.0.

- Graduate Merit Award, September 2022.
- Graduate Merit Award, February 2023.

Conference Papers

Sanmay Das, Fang-Yi Yu, and **Yuang Zhang**. “Optimally Auditing Adversarial Agents.” *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 40(20):16787–16794, 2026. DOI.

Fang-Yi Yu and **Yuang Zhang**. “Speakeasy: Auditing Cheap Signals in Peer Prediction.” Accepted to *NeurIPS 2026* (poster).

Workshop Papers

Yichi Zhang, Kevin Lu, **Yuang Zhang**, Jie Gao, Lirong Xia, and Fang-Yi Yu. “Mixture of Complementary Agents for Robust LLM Ensemble.” *NeurIPS 2025 Workshop on Socially Responsible and Trustworthy Foundation Models (ResponsibleFM)*, December 2025.

Research Experience

Doctoral Researcher, George Mason University

2023–present

Optimally Auditing Adversarial Agents

AAAI 2026

- Co-developed an ε -optimal audit algorithm running in $O(m^2)$ time for m private types by reducing continuous policy optimization to a finite search; accounted for agents’ worst-case equilibrium selection.
- Analyzed adaptive and budget-constrained extensions; simulated how audit costs, penalties, and type distributions affect misreporting, principal utility, and social welfare.

Speakeasy: Auditing Cheap Signals in Peer Prediction

NeurIPS 2026

- Co-developed a peer-prediction mechanism that penalizes agreement with sampled cheap signals; computed optimal audit distributions and characterized cost thresholds for truthful equilibria in the presence of correlated LLM outputs.
- Benchmarked five base mechanisms on ICLR peer-review data and HANS, demonstrating restored truth-telling incentives in peer review; used EXP3 and LLM agents to study truthful convergence and persistent coordination on cheap signals.

Truthful Data Procurement via Experimental Design (GRADES)

- Developing a mechanism that selects datasets from features before observing labels, combining distorted greedy allocation, critical-bid payments, and data quality bonuses to incentivize truthful cost reports and label delivery.
- Implementing benchmarks on Gaussian regression, MIMIC-III, RSNA Bone Age, Fitzpatrick17K, and DrugLib; comparing prediction error, welfare, acquisition cost, and selection runtime against DAVED, Data-OOB, and random selection.

Work in Progress

“Truthful Data Procurement via Experimental Design” (GRADES). Research manuscript, 2026.

Selected Presentations

“Optimally Auditing Adversarial Agents”: ACM Conference on Economics and Computation (EC), 2025 (poster); AAAI, 2026 (oral presentation).

Teaching

Graduate Teaching Assistant (three years), George Mason University *2023–present*

- **CS 262 (primary assignment)**: Led programming lab sessions, walked students through lab exercises, and helped troubleshoot their code.
- Held CS 262 office hours for programming assignments and course concepts; graded assignments and provided feedback on students’ solutions.
- **CS 692**: Held office hours for advanced-course homework, explained relevant technical concepts, and discussed research papers and students’ questions about their methods and arguments.

Teaching Assistant, The George Washington University *Aug. 2022–May 2023*

- Collaborated with other teaching assistants to write and peer-review homework solutions, checking their correctness and clarity.
- Graded student assignments and proctored midterm and final examinations.

Technical Skills

- **Programming and Scientific Computing**: Python (NumPy, SciPy, pandas).
- **Optimization and Statistical Modeling**: Convex optimization (CVXPY), least-squares regression, experimental design, and model evaluation (scikit-learn).
- **Computational Experiments**: Game-theoretic simulations, EXP3 learning dynamics, LLM-agent evaluation, benchmarking, and sensitivity analysis.
- **Research Tools**: Jupyter Notebook, Matplotlib, L^AT_EX, and Overleaf.